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7 February 1983
SUPERSEDING
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2 November 1977

MILITARY SPECIFICATION

COMPRESSOR UNIT, RECIPROCATING, GASOLINE-ENGINE-DRIVEN, AIR, 4-CFM, 0- TO 3,500-PSIG, WHEEL MOUNTED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 4 cubic feet per minute (cfm), 0- to 3,500 pounds per square inch, gage (psig), gasoline-engine-driven, wheel-mounted, air compressor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-N-290	- Nickel Plating (Electrodeposited).
QQ-P-416	- Plating, Cadmium (Electrodeposited).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4310

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MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3600	- Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-V-4719	- Valves, Oxygen Line, High Pressure, Aircraft Servicing.
MIL-M-8090	- Mobility, Towed Aerospace Ground Equipment, General Requirements for.
MIL-G-8348	- Gage Assemblies, Air pressure, Dial Indicating Chuck Type, Self Contained (Asg).
MIL-R-8573	- Reservoirs, Air, Nonshatterable Steel.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-14550	Copper plating (Electrodeposited).
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-25363	- Tank, Pneumatic Pressure, Aircraft, Glass Fiber.
MIL-C-26058	- Cartridge, Gas Purifier, Extremely Low Dewpoint, Type MA-2.
MIL-H-38390	- Hose Assembly, Tetrafluoroethylene, Pneumatic, High Pressure.
MIL-C-81751	- Coating, Metallic Ceramic.
MIL-C-83488	- Coating, Aluminum, Ion Vapor Deposited.
MIL-C-83959	- Cylinder, Air Purifier.
MIL-C-87115	- Coating, Immersion Zinc Flake/Chromate Dispersion.

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.

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MIL-STD-209	- Slinging and Tiedown Provisions For Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MS28889	- Valve, Air, High Pressure Charging.
MS33656	- Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
Ms49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.

DRAWINGS

AIR FORCE

7347114	- Chuck Assembly, High Pressure Air.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 85 - Control of Air Pollution From New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20404).

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations, Title 49, Chapter 1, Section 178.37 Specification 3AA; Seamless Steel Cylinders Made of Definitely Prescribed Steels.

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR (DOL)

Occupational Safety and Health Administration.
29CFR 1910 OSHA Safety and Health Standards.

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(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code.
Section VIII, Rules for Construction of Unfired Pressure Vessels.
Section IX, Welding Qualifications.

Performance Test Code PTC–9
Displacement Compressors, Vacuum, pumps, and Blowers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The compressor unit (hereinafter called "compressor") shall consist of a gasoline engine, fuel storage tank, pneumatic system with multistage reciprocating compressor pump, and all necessary accessories, assembled within a wheel-mounted frame, to produce a complete, self-contained portable unit. The dry weight of the complete compressor shall not exceed 300 pounds. The maximum dimensions shall be 55 inches long, 28 inches wide, and 35 inches high, including all attachments. The compressor shall withstand the operating and proof pressures specified herein without leakage or permanent deformation.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures) etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (reproduction model). The contractor shall furnish a compressor for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce compressors that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and, unless otherwise specified (see 6.2), all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.3 Material. Material shall be as specified herein and as shown on the drawing . Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7).

3.3.1 Corrosion-resistant treatment. Corrosion-resisting treatment shall consist of galvanizing to G90 coating designation for steel sheets or electrodepositing in accordance with QQ-P-416 Class 1, Type II or chromium in accordance with QQ-C-320 Class I, optional finish, or copper in accordance with MIL-C-14550, Class 1, or nickel in accordance with QQ-N-290, Class 1, Grade C. In addition the following coatings may be used: Metallic-ceramic in accordance with MIL-C-81751 (Type and class selection depends on application); aluminum in accordance with MIL-C-83488 Class 1, Type II and a dispersion coating in accordance with MIL-C-87115, Class 1.

3.3.2 Dissimilar metals. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-STD-889, shall be insulated from each other by a coat of chromate primer, zinc chromate tape, electrical tape, or other equally effective means.

3.3.3 Material deterioration prevention and control. The COmpreSSOr shall be fabricated from compatible materials that are inherently corrosion resistant or

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suitably treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments.

3.3.4 Identification of materials and finishes. The contractors shall have written documentation which identifies the specific material and material finish or treatment for use with components and subcomponents that make up a complete compressor unit. This documentation shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.4 Environmental conditions.

3.4.1 Storage temperatures. The compressor shall not be damaged by storage at a temperature of minus 50° F for periods up to 6 hours, and plus 155° F to 160° F for periods up to 4 hours.

3.4.2 Operation. The compressor shall operate as specified herein at any temperature from minus 25° F to plus 120° F.

3.4.3 Precipitation. The compressor shall perform as specified herein in rain falling at a rate of 3 inches per hour at any angle from the vertical to 45 degrees from the vertical. Water shall not leak into the air intakes or crankcases.

3.5 Safety. All parts which are subject to high operating temperatures and all moving parts shall be insulated, fully enclosed, or guarded and in compliance with OSHA regulations. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine or exhaust. Exhaust or discharges from the compressor shall be directed so that they do not endanger personnel, Protective devices shall not impair the operating functions. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Warnings shall be mounted on or near components containing hidden hazards such as automatic blow-down discharges.

3.5.1 Sound level. The sound level of the compressor while it is operating at full speed, shall be in accordance with MIL-STD-1474, Category D for steady-state noise when tested in accordance with 4.6.2.16. If the sound level of the compressor at the microphone location and conditions exceeds 85 dB(A), documentation that category D limits are clearly beyond the state-of-the-art shall be provided, noise levels will be reduced to the lowest feasible, and the distance from the compressor at which a limit of 85 dB(A) is met shall be determined. That distance shall be shown on hazard signs mounted on both sides of each production compressor, stating: “Caution hearing protection required within ____ feet when equipment is operating” (see 3.16). The caution shall be legible from the distance shown on the hazard sign. Hearing protectors need not be supplied with the compressor. When the caution notice is required, manuals shall include the appropriate discussion of the noise hazard in accordance with MIL-STD-1474, including the hearing protection, the” noise level of the item, and the distance from the compressor at which the 85 dB(A) limit is met.

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3.6 Maintainability.

3.6.1 Maintainability. The compressor shall operate as specified herein without repair or maintenance other than the manufacturer's recommended normal scheduled adjustments and servicing as established by a maintenance schedule prepared and submitted by the contractor prior to test or corrective maintenance actions. All major assemblies shall be accessible for servicing, repair, and replacement without the removal of other major assemblies. Cover plates which must be removed for component adjustment or repair shall be equipped with quick-disconnect fastenings. All fasteners shall be of corrosion-resistant material or shall be treated to be corrosion-resistant. All screw threads shall be in accordance with FED-STD-H28. Self-tapping screws shall not be used. Sizes and types of hardware and fastening devices shall be held to a minimum. Means for drainage of lubricants, fuel, and condensate traps shall be in accessible location and shall be accomplished without splashing on any component. All maintenance, assembly, or disassembly operations shall be accomplished with common tools, and special tools furnished with the compressor.

3.7 Performance.

3.7.1 Rated capacity. The rated capacity of the compressor shall be 4 cfm of free air at a delivery pressure of 3,500 psig under all conditions specified herein. The average capacity of any lot of 10 or more production compressors shall be not less than 4 cfm where no individual compressor shall produce less than 3.9 cfm or for computation purposes be credited for more than 4.2 cfm.

3.7.2 Tilted operation. When tilted in any position at an angle of 15 degrees from the horizontal, with the oil level in the engine and compressor on the "low" or "add" mark when checked with unit on level surface the compressor shall deliver the rated capacity for 30 minutes and shall not show evidence of leaks, faulty lubrication, inadequate fuel supply or lubricant spillover.

3.7.3 Operational life. The compressor shall operate as specified herein for 500 hours without overhaul, except that at the completion of 500 hours, air delivery at 3,500 psig shall be not less than 3.60 cfm, nor less than 90 percent of original delivery capacity, whichever is greater.

3.8 Pneumatic system. The pneumatic system shall consist of a compressor pump, valves, moisture separator, dehydrator, all necessary tubing and fittings, receiver, control panel, and servicing hose.

3.8.1 Compressor pump. The compressor pump shall be air-cooled, reciprocating, and single acting with not less than three stages of compression. A replaceable air intake filter with a nominal rating not exceeding 40 microns shall be provided. Intercoolers shall be provided between consecutive stages and an aftercooler shall be provided at the outlet of the last stage, to remove the heat of compression. The aftercooler shall be designed to cool the air to a temperature not greater than 30° F above ambient temperature.

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3.8.2 Relief valves. Regulation of the pneumatic system pressure downstream of the compressor pump shall be provided by a relief valve set to unseat at a pressure between 3,700 psig and 3,900 psig, and to reseat at a pressure not less than 3,500 psig. Additional relief valves shall be provided to individually protect the moisture separator, the air receiver, and each stage of the compressor pump, except that the last stage may be protected by a rupture disk. In addition to the relief valves, rupture disks shall be provided to protect the moisture separator and the air receiver. All relief valves shall be spring loaded, adjustable, preset, and securely locked and sealed to prevent unauthorized adjustment. Each relief valve and rupture disk shall be capable of venting full delivery capacity.

3.8.3 Moisture separator. A mechanical-type moisture separator shall be provided downstream of the aftercooler to remove condensed moisture and oil. The separator shall be designed for a minimum operating pressure of 3,500 psig and a proof pressure of 5,250 psig and shall be of a corrosion-resistant material. The separator shall be large enough to contain the moisture and oil which would accumulate in 20 minutes while the compressor is operating at rated capacity with a relative humidity of not less than 85 percent and an ambient temperature of 120″ F. A valve operated from the control panel shall be provided for periodic drainage of the moisture separator. Periodic drainage shall not be required more frequently than 15 minute intervals under all conditions specified herein. In addition, automatic drainage of the moisture separator shall occur on shutdown of the compressor. Provision shall be made to heat the moisture separator when operating at ambient temperature below 40° F to prevent freezing of condensed moisture, When an electric heater is utilized, it shall be designed for 28-volt d.c. electric service and the outlet for the service line shall be located on the control panel and shall mate with a plug conforming to MS3106A145-7S, with voltage applied across the A and B contacts.

3.8.4 Dehydrator. A chemical-type dehydrator shall be provided and shall be of the replaceable cartridge type, conforming to MIL-C-83959 and provided with an MA-2 cartridge conforming to MIL-C-26058. The cylinder shall be rigidly mounted in an accessible location and in any position from the vertical to 30 degrees from the vertical measured when the compressor is in a normal level position. Instructions for changing the cartridge in the cylinder shall be provided on an instruction plate (see 3.16). Instructions shall also include the required hours between cartridge changes for not less than three temperature ranges to maintain a delivery air dewpoint of minus 65° F as determined during testing. Interpolation and extrapolation of test results will be permitted. A blank space shall be provided on the instruction plate on which the hourmeter reading can be recorded. A line filter with a replaceable element rated at not greater than 10 microns shall be installed downstream of the dehydrator.

3.8.5 Back pressure valve. A back pressure valve shall be provided to maintain sufficient pressure to ensure the correct operation of the moisture separator and dehydrator.

3.8.6 Check valve. The pnematic system shall be equipped with a check valve to prevent reverse air flow from the receiver into upstream components.

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3.8.7 Pressure tubing and fittings. All pressure tubing and fittings shall be designed for a minimum operating pressure of 3,500 psig and a proof pressure of 5,250 psig and shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. All tubing and fittings shall be installed to minimize and withstand vibration and allow for thermal expansion. Provision shall be made for relieving pressure to the atmosphere from all lines in the system without disassembly of any fitting or component.

3.8,8 Receiver. A receiver designed for a minimum operating pressure of 3,500 psig and a proof pressure of 5,250 psig, with a capacity of not less than 1,300 cubic inches, shall be provided for storage of dried and filtered high-pressure air. The receiver shall conform to the requirements specified herein, and the design requirements of at least one of the following:

a. The ASME Rules for Construction of Unfired Pressure Vessels.
b. DOT Specification 3AA, for Seamless Steel Cylinder of Definitely Prescribed Steels.
c. MIL-T-25363, For Glass Fiber Tanks.
d. MIL-T-8573, For Nonshatterable Steel Cylinders.

The receiver shall be pad mounted and securely tied down. A piping system shall be provided to bypass the receiver for direct servicing of equipment. A provision shall be made to drain any accumulated moisture from the receiver. A pressure gage shall be provided to indicate receiver pressure. The gage shall be the oil fill type to control needle vibration and cover a range of O to 5,000 psig with 100 psig or smaller increments and shall be accurate to within plus or minus 1 percent of the full scale reading. The portion of the gage scale indicating pressures greater than 3,900 psig shall be marked with a red arc or sector.

3.8.9 Control panel. A control panel shall be provided for mounting the gages, meters, and controls specified herein. Gages, meters, and controls shall be clearly identified and shall be mounted so that they will not be damaged under any condition of compressor operation specified herein. All controls, gages, and meters necessary for operation shall be accessible and visible to a single operator standing on one side of the compressor.

3.8.9.1 Valves. Manually operated shutoff, isolation or control service valves shall be of the positive shutoff type and shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. The valves shall require not more than 60 inch-pounds of torque to open or close under a pressure of 3,500 psig and shall be designed so that overtorquing to 150 inch-pounds will not damage the valve seats.

3.8.9.2 Regulator. Unless otherwise specified (see 6.2), a single high-low pressure regulator shall be provided and mounted on the control panel. The regulator shall be designed for an operating pressure of 3,500 psig and a proof pressure of 5,250 psig. The regulator shall be adjustable to any outlet pressure setting between 0 psig and 3,500 psig by means of a single-control. Not more

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than 50 inch-pounds of torque shall be required to operate the control. By counterclockwise rotation of the control, the regulator shall bleed the downstream line to any desired lower pressure including atmospheric without loss of upstream air. The regulator shall control the downstream pressure without variation from a preset pressure by more than plus or minus 2 percent when the upstream pressure is decreased. A differential-type overpressure relief device shall be provided on the downstram side of the regulator. The relief device shall protect the downstream side in the event of failure of the regulator. The relief device shall start to vent before the downstream pressure reaches 100 psi above any set dead end pressure up to 3,500 psig, and flow shall reach at least 10 cfm before the downstream pressure reaches 200 psi above the set dead end pressure. A filter shall be provided at the inlet of the regulator to prevent foreign particles from entering the regulator. Two pressure gages, with a minimum diameter of 3 inches, shall be provided on the downstream side of the pressure regulator and shall be mounted on the control panel. The high-pressure gage shall cover the range of O to 3,500 psig with increments of not more than 50 psi and the low-pressure gage shall cover the range of O to 600 psig with increments of not more than 10 psi. Provision shall be made to protect the low pressure gage aginst damage when pressure is in excess of the full scale reading. Gages shall be accurate to within plus or minus 1 percent of full scale. A manually operated shutoff valve shall be provided downstream of the regulator and relief valve and shall be mounted on the control panel.

3.8.9.3 Hourmeter. An hourmeter shall be mounted on the control panel. The hourmeter shall have a range of from 0 to 9,999 and an accuracy of 1 minute per hour. The hourmeter may be of the revolution-counter type.

3.8.10 Servicing hose. A servicing nose conforming to MIL-H-38390 with couplings to mate with fittings conforming to MS33656-4, shall be provided. Unless otherwise specified (see 6.2), the hose shall be not less than 20 feet in length and shall be equipped with coupling guards to minimize hose failure due to bending at or near the couplings. When specified, a remote control screw-on gage assembly conforming to MIL-G-8348, Class C shall be furnished, with the exception that the body shall be constructed to withstand a maximum working pressure of 3,500 psig and a minimum bursting pressure of 5,250 psig. A servicing line valve conforming to MIL-V-4719, Type III shall be provided at one end of the hose. A 9-inch-long hose extension shall be provided and attached to the servicing line valve. Unless otherwise specified (see 6.2), a screw-on-type air chuck designed to fit air valves conforming to MS28889 shall be attached to the hose extension. The chuck shall conform to Drawing 7347114 except the material may be brass or stainless steel, and shall be provided with a dust cap or dust plug with chain to prevent loss. The hose shall be provided with a quick-disconnect fitting designed for an operating pressure of 3,500 psig to couple with the terminal fitting of the pneumatic system mounted on the control panel. The coupling shall be a two way shut-off type and shall be a Hansen Series 2 HK or equal. It shall be constructed of corrosion resistant steel or cadmium plated in accordance with QQ-P-416, Type I, Class 3. A rack or storage area shall be provided for storage of the hose assembly. The minimum bend radius of the hose shall not be exceeded during storage and provision shall be made to secure the line service valve if rack mounted. Provision shall be made to prevent uncoiling of the hose during transportation and storage.

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3.9 Engine. For the purpose of selecting the engine, the compressor shall be classified as a class II end item in accordance with MI1-STD-141O. The gasoline engine, including all systems, components and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. The engine shall be air cooled. The maximum governed speed for each engine, as applied, shall not exceed the engine manufacturer's recommended speed for this application. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the Environmental Protection Agency regulations governing control of air pollution for new motor vehicles and new engines, in effect for this type of equipment on date of manufacture. The engine shall meet all operational requirements and conditions specified herein without exceeding the limits specified in MIL-STD-1410, or, the engine manufacturer's recommended operating limits. The average oil consumption of each engine shall not exceed .0035 pound per brake-horsepower-hour (bhp/hr) when tested as specified herein.

3.9.1 Fuel tank. A fuel tank with a capacity for not less than 4 hours of operation at rated capacity shall be provided. Construction and mounting shall be such that vibration or excessive tightening of the mounting devices will not cause leaks to develop. The filler opening shall be a minimum of 1.5 inches in diameter, shall incorporate a removable strainer, and shall have a chain-secured filler cap. The fuel tank outlet shall be not less than 1/4 inch above the bottom of the fuel tank. The tank shall be located to minimize fire hazard or damage resulting from spilling, overflow, or drainage of fuel. When specified (see 6.2), provision shall be made for connection to an auxiliary fuel supply. The fuel tank shall be vented and shall not leak when tilted as specified in 3.7.2.

3.9.2 Engine speed control. The engine shall be furnished with a governing system conforming to MIL-STD-1410. A hand throttle shall be furnished and shall be capable of controlling the engine speed at any speed between the engine manufacturer's recommended low idle speed and the maximum governed speed.

3.9.3 Cranking system. An automatic rewind rope starter conforming to MIL-STD-1410 shall be furnished.

3,9.4 Air induction system. A heavy duty air cleaner and restriction indicator, in accordance with MIL-STD-1410, shall be furnished.

3.9.5 Exhaust system. The exhaust system shall be designed to present minimum hazard to the operator while starting, operating, or making minor adjustments to the engine or compressor unit while running. The system shall include a muffler and a spark arrester, or a combination muffler and spark arrester. The spark arrester shall conform to SAE J335. The total backpressure of the exhaust system shall not exceed the maximum recommended by the engine manufacturer.

3.10 Unit mounting. The compressor pump, engine, and all other components shall be mounted in and on an integral frame and chassis assembly. Provision shall be made to protect the compressor pump, high pressure piping, and

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accessories from damage due to vibration during operation or accidental damage from dropped objects. A two-wheel running gear shall be provided, designed and constructed in accordance with MIL-M-8090, Type I, Group A, Class 1, using tires not smaller than 3.50 by 6. Provisions of MIL-M-8090 concerning the following shall not apply to the compressor:

a. Towing provisions.
b. Brakes.
c. Landing gear.
d. Pintle hook, rear.
e. Reflectors.

When specified (see 6.2), provision shall be made for a towbar and lunette in accordance with MIL-M-8090, Type I, Group A, Class 2, except that dimensions and material may be altered to meet the light weight requirements of the unit. A folding or retractable handle shall be provided to permit easy movement of the compressor by one man. A mechanical parking brake and retractable landing gear shall be provided.

3.11 Electromagnetic interference. The electromagnetic interference emission characteristics of the compressor shall conform to MIL-STD-461, Class IIIC.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.12.1 Lubricants. The compressor shall operate as specified herein when lubricated with military lubricants (see 6.3).

3.12.2 Lubrication-fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.12.3 Pressure-release device. A pressure release device shall be provided where the use of pressure lubricating equivalent will damage grease seals or other parts.

3.12.4 Filling, draining, and checking provisions. Enclosures that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than l/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drain plug, size conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be accomplished when the compressor is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container.

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3.12.5 Grease lubrication. All grease lubrication including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
b. MIL-G-10924 - For bearings where operating temperatures so not exceed 175° F.

The compressor shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.12.6 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range as specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricants used. The oil used in the compressor pump shall conform to MIL-L-23699. The compressor pump shall have a self-contained lubricating system that shall not require addition of oil more often than every 50 hours accumulated operating time. Maximum oil consumption shall be 10 cubic centimeters (cc) per hour of operation.

3.13 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.13.1 with 7 percent salicylanilide or 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.13.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102 South Coast Air Quality Management District.

3.14 Slinging and tiedown provisions. Permanently affixed attachments for lifting and tying down the compressor shall be provided, and shall conform to MIL-STD-209 for type II equipment.

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3.15 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type 1, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.16 Instruction plates. Each compressor shall be equipped with instruction plates, including diagrams, warnings and cautions ascribing any special or important procedures to be follwed in assembling, operating, or servicing the compressor. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.16.1 Flow diagram. Each compressor shall be provided with a schematic flow diagram of the pneumatic system. The diagram shall either be part of the control panel, with all gages, meters, and controls located in place on the diagram, or shall be located near the control panel in such a way that it is visible to an operator utilizing the controls.

3.16.2 Operating instructions. Operating instructions shall be located on or near the control panel so that they are visible to an operator utilizing the controls. Instructions shall be clear and concise and shall provide sufficient information to enable operation without injury to personnel or damage to the compressor.

3.16.3 Caution notice. A notice shall be mounted adjacent to the terminal fitting of the pneumatic system stating:

CAUTION
HIGH PRESSURE AIR

3.17 Treatment and painting. The portions of the compressor normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, lusterless, color as specified (see 6.2).

3.18 Workmanship.

3.18.1 Castings and forgings. All parts, components, and assemblies of the compressor which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.18.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.18.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter other than mill scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Unless otherwise specified, the welding process used in fabrication of the compressor shall be at the option of the contractor.

3.18.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of weld operations to be performed and that such qualification is effective as defined by the particular code:

Welding Qualification of the ASME.
AWS, Structural Welding Code, Section 5, Qualification, D1-1.

Contractors who only make horizontal welds need not qualify welders for “all position welding”. Subject to approval by the Government, contractor’s standard welder qualification may be substituted in lieu of the above codes provided that the contractor’s procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.18.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.18.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contrator may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c, Inspection of packaging (see 4.7).

4,3 Preproduction inspection.

4.3.1 Examination. The compressor shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The compressor shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection.

4.4 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examination specified in table I, column 1 and the tests specified in table II, column 1 to determine that the quality of the selected compressor is equal to the quality standards established during preproduction inspection. The inspection will be performed by the Government at a site selected by the Government. Compressors will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model.

4.4.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each compressor shall be examined for the defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each compressor shall be subjected to the tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examinations shall be in accordance with table I.

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TABLE I. Examination Schedule.

Preproduction	Quality Conformance	Defect	Requirement Paragraph(s)
1	2	3	4
X	-	101. Dimensions and weight not as specified.	3.1
X	-	102. Materials not as specified.	3.3, 3.3.3, 3.8.7, 3.8.9.1
X	-	103. Corrosion resistant treatment not as specified.	3.3.1
X	-	104. Presence of non-permissible galvanic couples.	3.3.2, 3.3.3
X	-	105. Safety provisions not as specified	3.5
X	-	106. Maintainability provisions not as specified.	3.6.1
X	-	107. Pneumatic system not as specified.	3.8
X	-	108. Compressor pump design not as specified.	3.8.1
X	X	109. Relief valves missing or not as specified.	3.8.2
X	X	110. Moisture separator missing or not as specified.	3.8.3
X	X	111. Dehydrator or line filter missing or not as specified.	3.8.4
X	X	112. Dehydrator cartridge missing.	3.8.4
X	X	113. Back pressure valve missing or not as specified.	3.8.5
X	X	114. Check valve missing or not as specified.	3.8.6
X	X	115. Tubing and fittings not as specified.	3.8.7
X	-	116. Receiver design and installation not as specified.	3.8.8
X	X	117. Gages missing or not as specified.	3.8.8, 3.8.9.2
X	-	118. Control panel not as specified.	3.8.9
X	-	119. Valves not as specified.	3.8.9.1
X	X	120. Regulator installation not as specified.	3.8.9.2
X	X	121. Hourmeter missing or not as specified.	3.8.9.3
X	X	122. Servicing hose and accessories missing or not as specified.	3.8.10

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TABLE I. Examination Schedule. (continued)

Preproduction	Quality Conformance	Defect	Requirement Paragraph(s)
1	2	3	4
X	-	123. Engine not as specified.	3.9, 3.9.3, 3.9.4
X	-	124. Fuel tank construction and mounting not as specified.	3.9.1
X	-	125. Unit mounting not as specified.	3.10
X	X	126. Lubrication provisions not as specified.	3.12 thru 3.12.6
X	X	127. Fungus and moisture resistance not as specified.	3.13
X	-	128. Slinging and tiedown provisions not as specified.	3.14
X	X	129. Identification marking not as specified.	3.15
X	X	130. Instruction plates missing or not as specified.	3.16 thru 3.16.3
X	X	131. Treatment and painting not as specified.	3.17
X	X	132. Workmanship not as specified	3.18 thru 3.18.5
X	-	133. Contractor not having documentation for identification of material finishes.	3.3.4

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to test, the compressor shall be lubricated with oils and greases specified in 3.12.1, 3.12.5, and 3.12.6. Oils shall be those designated for use in the ambient temperature at the place of test.

4.6.2.2 Test schedule. Tests shall be in accordance with table II.

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TABLE II. Test Schedule.

Preproduction	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	X	Test conditions.	4.6.2.1	3.12.1, 3.12.5 and 3.12.6
X	-	Power consumption.	4.6.2.3	3.9.1
X	-	Hydrostatic pressure.	4.6.2.4	3.8.7
		Pressure regulator tests:	4.6.2.5	-
X	-	Hydrostatic.	4.6.2.5.1	3.8.9.2
X	-	Pressure range.	4.6.2.5.2	3.8.9.2
X	-	Outlet pressure variation.	4.6.2.5.3	3.8.9.2
X	-	Overpressure relief device	4.6.2.5.4	3.8.9.2
X	-	Gage accuracy.	4.6.2.6	3.8.8 and 3.8.9.2
X	-	Initial performance.	4.6.2.7.1	3.7.1, 3.8.1, 3.8.4 and 3.12.6
X	-	Moisture separator.	4.6.2.8	3.8.3
X	-	High temperature.	4.6.2.9	3.4.1 and 3.4.2
X	-	Low temperature.	4.6.2.10	3.4.1 and 3.4.2
X	-	Precipitation.	4.6.2.11	3.4.3
X	-	Tilted position.	4.6.2.12	3.7.2
X	-	Slinging and tiedown provisions.	4.6.2.13	3.14
X	-	Electromagnetic interference.	4.6.2.14	3.11
X	-	Endurance	4.6.2.15	3.6, 3.7.1, 3.7.3, 3.8.1, 3.8.2, 3.8.4, 3.8.9.1, 3.9.2, 3.12.6
X	-	Sound level.	4.6.2.16	3.5.1
-	X	Operational run-in	4.6.2.17	3.7.1

4.6.2.3 Power consumption. Using a dynamometer or a calibrated motor, operate the compressor pump at rated capacity and speed, and determine the brake horsepower required. A power requirement in excess of the engine manufacturers advertised continuous brake horsepower rating at the applicable speed shall constitute failure of this test.

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4.6.2.4 Hydrostatic pressure. Subject all pressure vessels, tubing and fittings (except pressure gages and relief devices) to a hydrostatic pressure of 5,250 psig and hold at this pressure until examined for leaks and distortion. Any evidence of leakage or permanent distortion shall constitute failure of this test.

4.6.2.5 Pressure regulator tests.

4.6.2.5.1 Hydrostatic. Subject the high- and low-pressure zones of the regulator to the test specified in 4.6.2.4. After hydrostatic testing, disassemble and inspect the regulator for damage. Any evidence of leakage or any indication of permanent deformation to any internal or external component shall constitute failure of this test.

4.6.2.5.2 Pressure range. With an inlet pressure of 3,500 psig, set the regulator for O psig outlet pressure. With the outlet closed and using a device to measure torque, increase the outlet pressure to 500 psig in steps of 25 psi, then to 3,500 psig in steps of 250 psi, and then reduce the outlet pressure to O psig using the same increments of pressure. A torque required to change the setting in excess of 50 inch-pounds, or failure to regulate the outlet pressure to any pressure between O psig and 3,500 psig inclusive, shall constitute failure of this test.

4,6.2.5.3 Outlet pressure variation. With an inlet pressure of 3,500 psig to the regulator, set the outlet pressure to 500 psig with a flow rate of 4 cfm. Without further adjustment slowly reduce the inlet pressure to 800 psig. An outlet pressure variation of more than plus or minus 2 percent shall constitute failure of this test.

4.6.2.5.4 Overpressure relief device. With a 3,500 psig inlet pressure, adjust the regulator for a 500 psig outlet pressure. Without further adjustment of the regulator, connect a variable source of higher pressure gas to the regulator outlet. Note the pressure at which the pressure relief device starts to vent and measure the flow rate of the effluent gas as the outlet pressure is increased. Lower the pressure on the outlet and note the pressure at which the pressure relief device reseats. An initial start-to-vent pressure of more than 600 psig, an effluent gas flow rate of less than 10 cfm at 700 psig, or a reseating pressure of less than 500 psig shall constitute failure of this test.

4.6.2.6 Gage accuracy. Dead weight test the receiver and regulator pressure gages throughout the pressure range increments no greater than 10 percent of the full-scale gage reading. Any reading having an error greater than 1 percent of the full-scale reading of the gage shall constitute failure of this test.

4.6.2.7 Initial performance.

4.6.2.7.1 Test procedure. Operate the compressor with the regulator fully open, and the back pressure on the outlet maintained at 3,500 psig, until the specified temperature data have stabilized; then continue operation for a period of not less than 2 hours, recording data in 10-minute intervals. The data required are:

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a. Temperature of air at receiver outlet, degrees Fahrenheit.
b. Aftercooler inlet and outlet temperatures, degrees Fahrenheit.
c. Temperature air intake, degrees Fahrenheit.
d. Ambient air temperature and pressure, degrees Fahrenheit and inches Hg.
e. Lubricating oil temperatures, degrees Fahrenheit.
f. Oil consumption (record only at end of 2-hour run).
g. Dewpoint of delivered air, degrees Fahrenheit, measured by a coulometric, electrolytic-type, dial-indicating hygrometer.
h. Compressor crankshaft speed, rpm.
i. Discharge pressure, psi.
j. Air delivery rate, cfm, determined in accordance with the ASME Performance Test Code PTC-9, utilizing a long-radius flow nozzle; or by calculation of a weight of air delivered in a given time against a back pressure of 3,500 psig, wherein the volume of the pressure vessel utilized for weight measurement shall not be less than 1,000 cubic inches and capacity shall be calculated in terms of free air at intake conditions.

4.6.2.7.2 Failure criteria. Any one or more of the following shall constitute failure of this test:

a. Aftercooler outlet temperature greater than 30° F above ambient temperature.
b. Oil consumption in excess of 10 cc per hour.
c. Dewpoint greater than minus 65° F.
d. Air delivery less than 4 cfm of free air at 3,500 psig.

4.6.2.8 Moisture separator. Operate the compressor for a minimum of 15 minutes at a temperature of 120° F and a relative humidity of 85 percent. (This test may be run in conjunction with the high-temperature test.) Examine the dehydrator and the lines connecting the moisture separator to the dehydrator for evidence of oil or water. Any condensed water or oil carryover from the separator into the dehydrator shall constitute failure of this test.

4.6.2.9 High temperature. Subject the compressor to a minimum ambient temperature of 155° F for a period of 4 hours, including 1 hour at not less than 160° F and examine for damage. Subject the compressor to an ambient temperature of 120° F plus or minus 5° F until the temperature data specified in 4.6.2.7.1 have stabilized. Start the compressor and operate at rated capacity for not less than 8 hours while maintaining 120° F plus or minus 5° F ambient temperature, and record the data specified in 4.6.2.7.1 at 30-minute intervals. Damage to the compressor, or any of the failure criteria specified in 4.6.2.7.2 shall constitute failure of this test.

4.6.2.10 Low temperature. Subject the compressor to an ambient temperature no greater than minus 50° F for a period of 6 hours without solar radiation, and examine for damage. Subject the compressor to an ambient temperature of minus 25° F plus or minus 5° F until the temperature data specified in 4.6.2.7.1 have stabilized (the moisture separator may be preheated). Start the compressor and

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operate at rated capacity for less than 4 hours while maintaining minus 25° F plus or minus 5° F ambient temperature, and record the data specified in 4.6.2.7.1 at 30-minute intervals. Damage to the compressor, or any of the failure criteria specified in 4.6.2.7.2 shall constitute failure of this test.

4.6.2.11 Precipitation. Subject the compressor, in operating condition with all panels necessary for operation open, to a simulated rainstorm by spraying fresh water at a minimum rate of 3 inches per hour directed simultaneously at all four sides and on the top at an angle of approximately 45 degrees from the vertical. The apparatus for producing the spray shall emit water in the form of droplets rather than a fine mist. Expose the compressor to the spray for 1/2 hour while not running and then start and operate the compressor for 1/2 hour with the outlet closed and the receiver bypassed while the spray continues. Water leakage into air cleaners and crankcases or water accumulation on other components which prevents normal operation shall constitute failure of this test.

4.6.2.12 Tilted position. Tilt the compressor in each of four positions 15 degrees from the horizontal plane about its transverse and longitudinal axes. Start the compressor in each of the four positions with a full fuel tank and operate for 30 minutes at rated capacity. Allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel system. Without changing the tilt, start the compressor and operate for 30 minutes at rated capacity. Evidence of leaks, faulty lubrication, inadequate fuel supply or lubricant spillover shall constitute failure of this test.

4.6.2.13 Slinging and tiedown provisions. The slinging and tiedown provisions shall be subjected to the loads specified in MIL-STD-209 to determine conformance with the slinging provision working load and tiedown eye ultimate strength (static load) requirements. Nonconformance with MIL-STD-209 shall constitute failure of this test.

4.6.2.14 Electromagnetic interference. The compressor shall be subjected to the tests specified in MIL-STD-461 for Class IIIC equipment. The test report covering electromagnetic interference shall be included in the final compressor test report. Nonconformance of the compressor with MIL-STD-461 shall constitute failure of this test.

4.6.2.15 Endurance.

4.6.2.15.1 Test procedure. Operate the compressor for 500 hours as follows:

a. 400 hours delivering a minimum of rated capacity.
b. 50 hours cycling a minimum of 10 times per hour between cut-in and cut-out pressures of the regulator.
c. 50 hours cycling once each hour, delivering rated capacity for a 10 minute period and delivering no air for a 50 minute period each hour.

Conduct the test without shelter and at the climatic conditions existing at the place of test. Record the data specified in 4.6.2.7.1 at 8-hour intervals. Readings shall be taken only after at least 1 hour of continuous operation.

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Endurance test time will not be credited in less than 2-hour intervals of continuous operation. Intervals of endurance test time of less than 4 hours terminated by malfunction will not be credited. All test time accrued during other successfully completed tests specified herein shall be credited. Operate all functional controls including valves, regulators, and control devices not less than 50 times during the test. All safety relief devices shall be caused to operate at least one time. Operation of safety devices may be demonstrated independently of compressor operation, Determine intervals of dehydrator cartridge change as specified in 3.8.4.

4.6.2.15.2 Failure criteria. Any one or more of the following shall constitute failure of this test:

a. Aftercooler outlet temperature greater than 30° F above ambient temperature.
b. Oil consumption in excess of 10 cc per hour.
c. Dew point greater than minus 65° F.
d. Air delivery during or at the conclusion of the test less than 3.60 cfm or less than 90 percent of original delivery capacity, at 3,500 psig, whichever is greater.
e. Valve operation not as specified in 3.8.2 and 3.8.9.1.
f. Fuel tank capacity insufficient for 4 hours of operation at rated capacity.
g. Failure or malfunction which is not corrected during normal maintenance as established by the contractor's maintenance schedule.

4.6.2.16 Noise. Place the compressor in a test area which is free of any large reflecting surfaces within 50 feet of the compressor or sound-level measuring equipment. With the compressor operated at rated load, a sound level survey shall be made around the compressor to determine the direction in which the maximum sound levels are propagated so that the test equipment can be properly oriented. The recording microphones shall be positioned in accordance with MIL-STD-1474, Category D. The noise level shall be measured with the compressor operated at no load and rated load. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.6.2.17 Operational run-in. Operate each compressor for not less than 12 hours utilizing all operator controls and determine capacity. Capacity may be measured by use of a flowmeter. An average capacity for any lot of 10 or more compressors of less than 4 cfm where no individual compressor shall produce less than 3.9 cfm or is credited for more than 4.2 cfm for the purpose of computing the average, or any malfunction, leakage, rough, irregular operation shall constitute failure of this test.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-C-3600.

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5. PACKAGING

5.1 Preservation, packing, and marking. Each complete compressor shall be preserved, packed, and marked in accordance with MIL-C-3600. The degree of preservation shall be level A, B, or commercial and the degree of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The compressor is intended to provide clean, dry, low- and high-pressure air for ground support of aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of the preproduction model (see 3.2).
d. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which will be conducted by the Government and which shall be conducted by the contractor (see 3.2)
e. When other than a single regulator is required (see 3.8.9.2).
f. When other than a 20-foot length of hose is required (see 3.8.10).
g. Whether air chuck is not required and when remote control gage assembly is required (see 3.8.10).
h. When provision shall be made for connection to an auxiliary fuel supply (see 3.9.1).
i. When provision shall be made for a towbar and lunette (see 3.10).
j. Color (see 3.17).
k. Degree of preservation and packing required (see 5.1).

6.3 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the compressor covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.4 Preproduction model. Any changes or deviations of production compressors from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preprodution model will not relieve the contractor of his obligation to furnish compressors conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor’s maintenance and operation manual to be furnished with each compressor.

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6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compressors. A suggested paragraph is as follows:

> "Shipment of compressors shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.8 International standardization. Certain provisions (see 3.7.1, 3.8.3 and 3.8.10) of this specification are the subject of International Standardization Agreement STANAG 3054 and ASCC AIR STD 14/7. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization offices that appropriate action may be taken respecting the international agreement concerned.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - AV

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